

Task 2)

Share an S3 bucket with another AWS account using an IAM Role so that only the IAM user with sts:AssumeRole permission can assume the role and access the S3 bucket.

Or

Share an S3 bucket from **Deep Meshram** account with **Kartik Deogade** account using an IAM Role. In Kartik's account, **user1 should be able to assume the role** and access the S3 bucket, but **user2 should not be able to assume the role**.

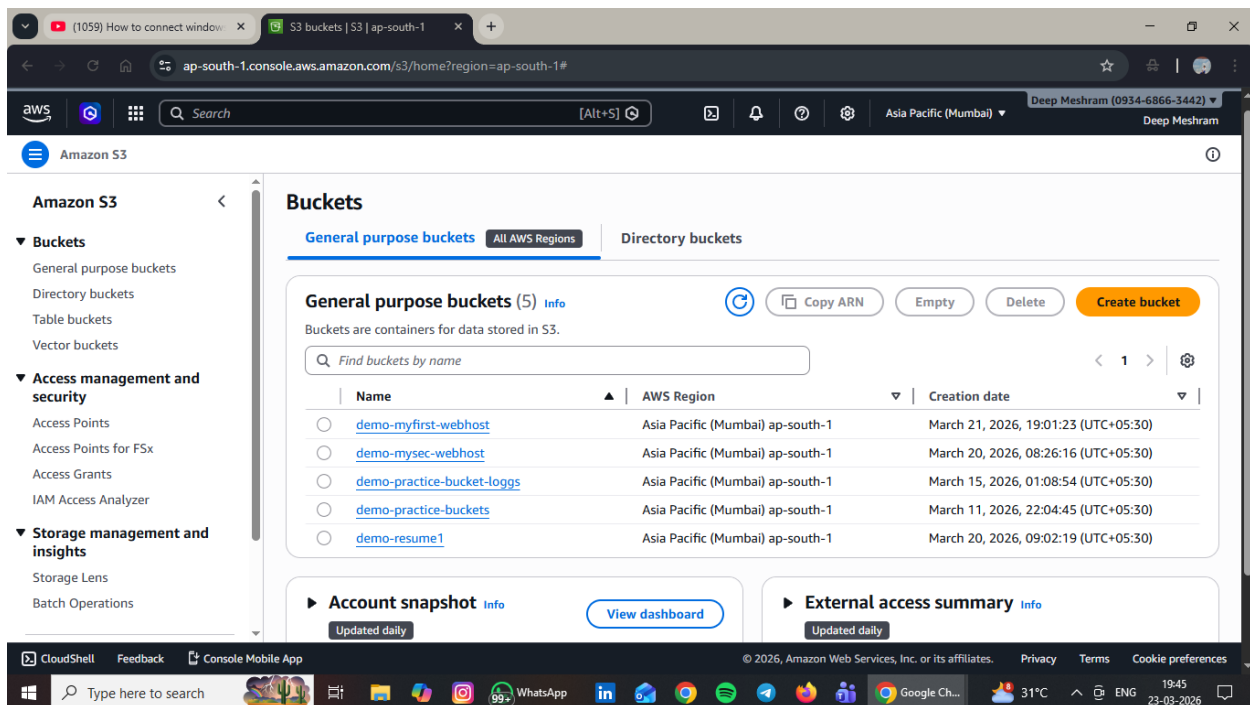
Account A – Deep Meshram (Bucket Owner Account) i.e Trusting account

Account B- Kartik Deogade (Trusted account)

From Account A

Step 1: Create S3 Bucket

Go to S3 → Create Bucket, enter bucket name and create the bucket.



Step 2: Create IAM Role for Kartik Account

Go to IAM → Roles → Create Role → Another AWS Account → Enter Kartik Deogade Account ID → Click Next.

The screenshot shows the AWS IAM console 'Create role' wizard. The browser address bar is `us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/roles/create`. The user is logged in as 'Deep Meshram (0934-6866-3442)'. The left sidebar shows the navigation menu with 'IAM' > 'Roles' > 'Create role'. The main content area is titled 'Select trusted entity' and includes a progress indicator on the left with three steps: 'Step 1: Select trusted entity' (active), 'Step 2: Add permissions', and 'Step 3: Name, review, and create'. Under 'Trusted entity type', there are five options: 'AWS service' (disabled), 'AWS account' (selected), 'Web identity' (disabled), 'SAML 2.0 federation' (disabled), and 'Custom trust policy' (disabled). Below this, the 'An AWS account' section is expanded, showing the option 'Another AWS account' selected. The 'Account ID' field contains the value '049930840693'. The 'Options' section has two checkboxes: 'Require external ID' (unchecked) and 'Require MFA' (unchecked). At the bottom right, there are 'Cancel' and 'Next' buttons.

Step 1
● Select trusted entity
○ Step 2: Add permissions
○ Step 3: Name, review, and create

Select trusted entity

Trusted entity type

- ☐ AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☒ AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

An AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

- ☐ This account (093468663442)
- ☒ Another AWS account

Account ID
Identifier of the account that can use this role

049930840693

Account ID is a 12-digit number.

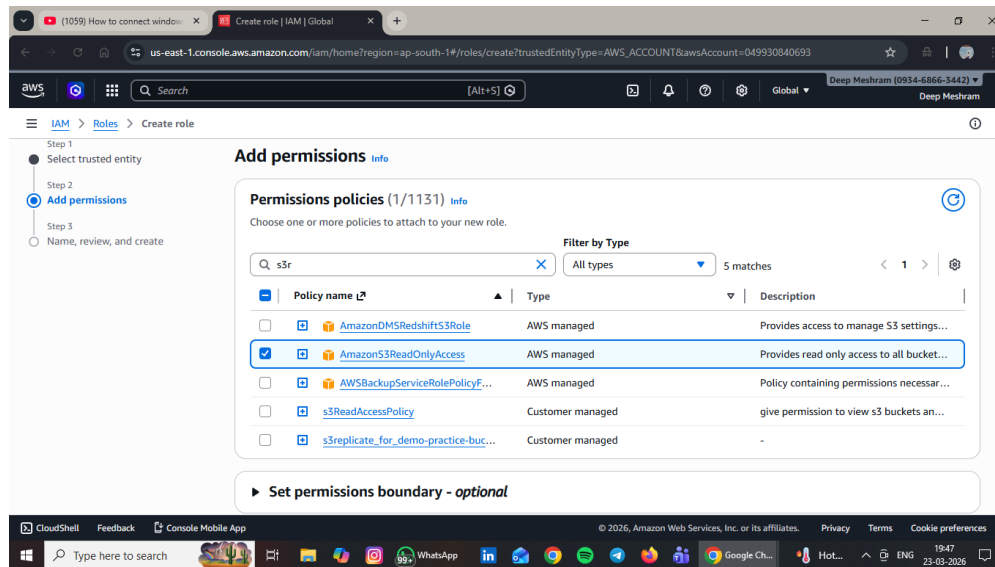
Options

- ☐ Require external ID (Best practice when a third party will assume this role)
- ☐ Require MFA
Requires that the assuming entity use multi-factor authentication.

Cancel Next

Step 3: Attach S3 Permission Policy to Role

Attach a custom policy that allows AmazonS3ReadOnlyAccess

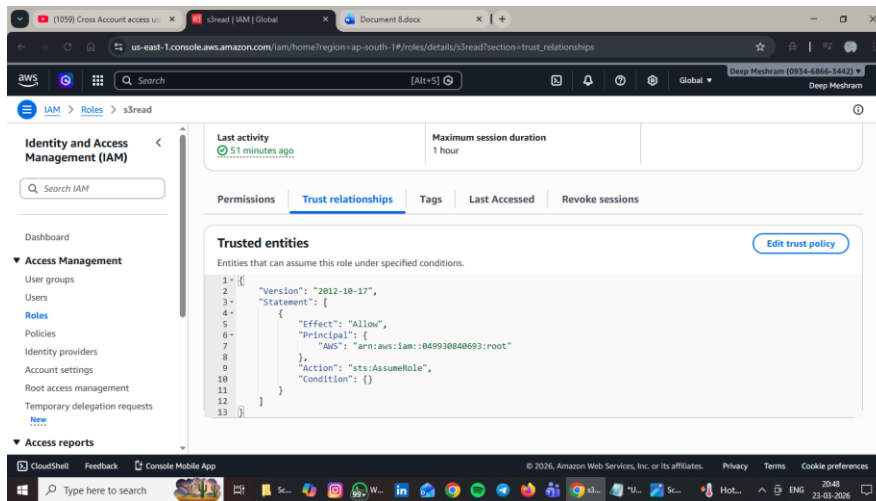


Step 4: Check Trust Relationship

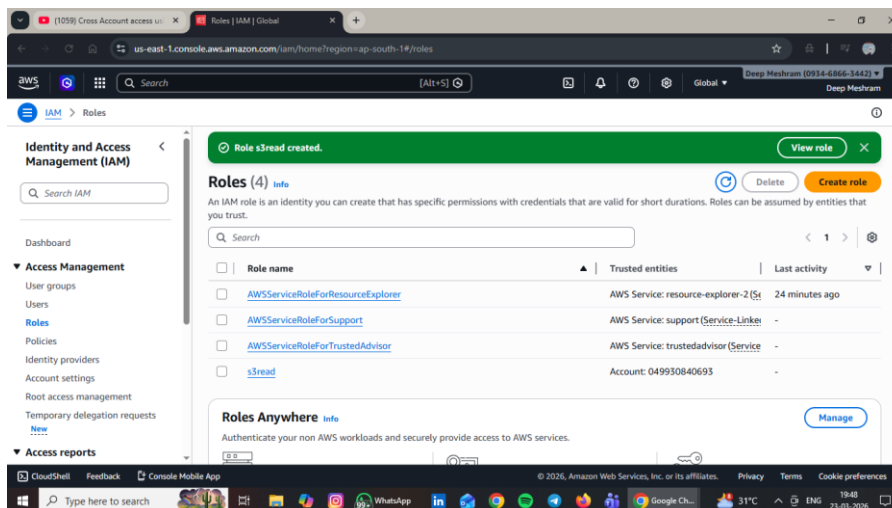
Go to Role → Trust Relationships → Edit Trust Policy and add Kartik account.

JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::KARTIK-ACCOUNT-ID:root"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```



Step 5: create role with name s3read



Now From Account B – Kartik Deogade (Trusted Account)

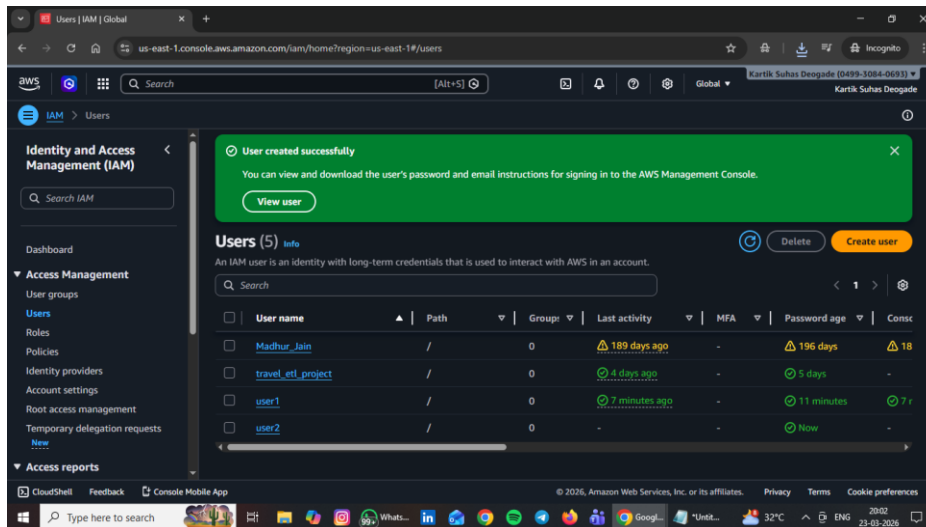
Step 6: Create IAM Users

Create two IAM users:

user1

user2

Go to IAM → Users → Create User.



Step 7: Give sts:AssumeRole Permission to user1 Only

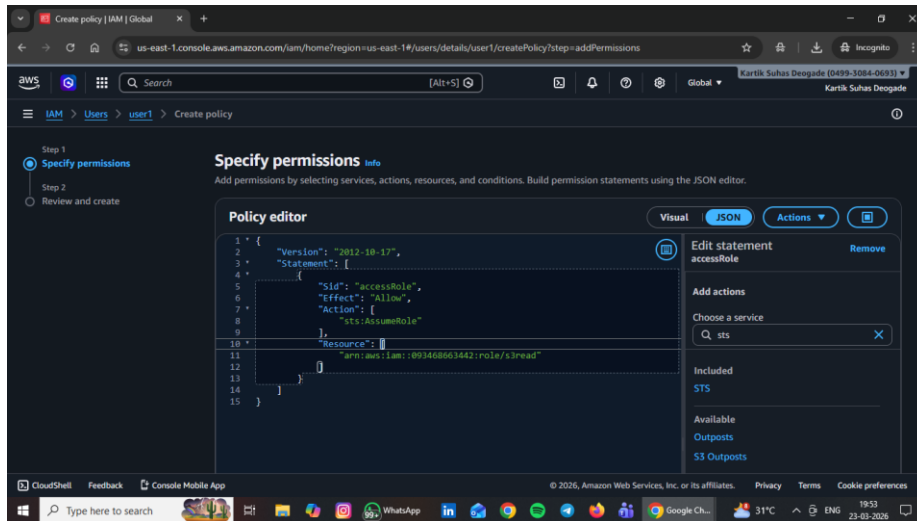
Go to IAM → user1 → Add Inline Policy → JSON and add:

JSON

```
{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Effect": "Allow",  
      "Action": "sts:AssumeRole",  
      "Resource": "arn:aws:iam::DEEP-ACCOUNT-ID:role/ROLE-NAME"
```

```
}  
]  
}
```

Note: user2 does NOT get this permission.



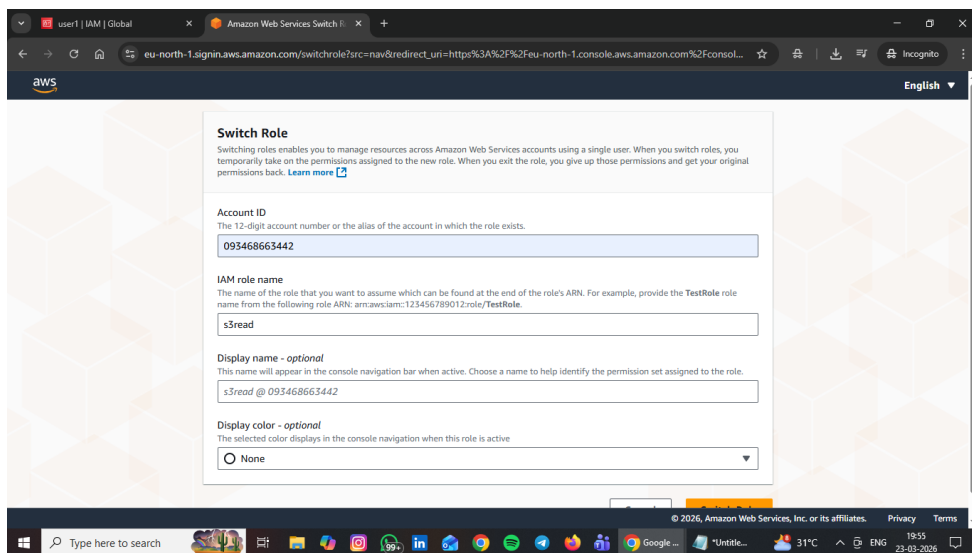
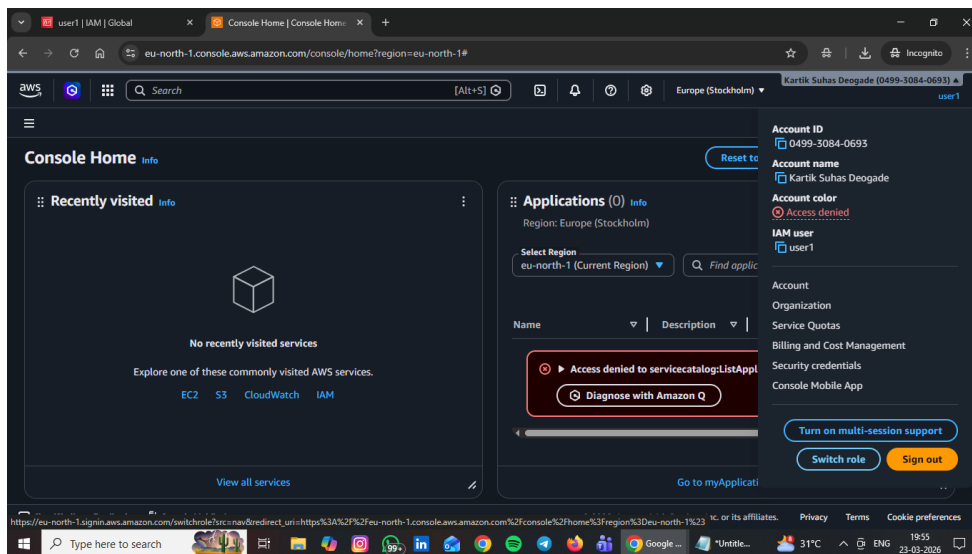
Step 8: Switch Role Using user1

Login as user1 → Click Switch Role → Enter:

Account ID → Deep Meshram Account ID

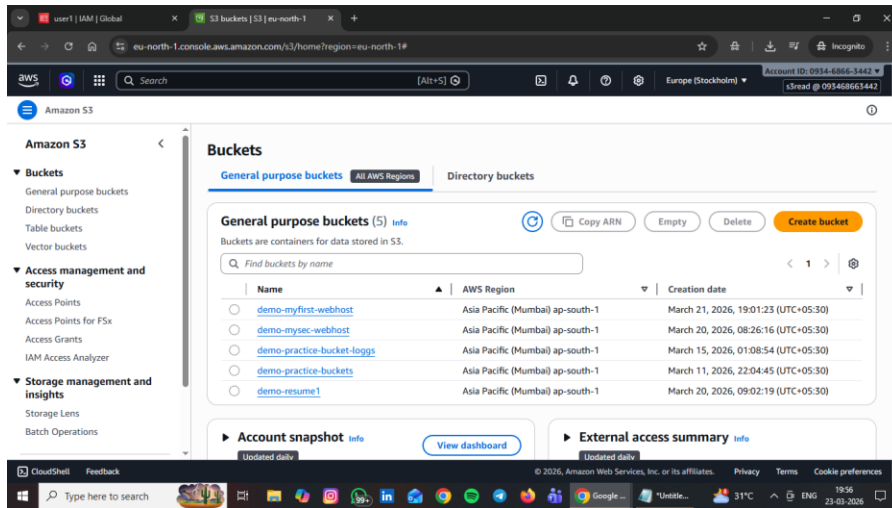
Role Name → Role created in Account A

Display Name → Any name → Click Switch Role



Step 9: Verify S3 Access (user1)

After switching role → Go to S3 → Bucket → Objects → user1 can list and download objects.



Step 10: Verify Switch Role from user2 (Should Fail)

Login as user2 → Try Switch Role → Enter Account ID and Role Name →

Switch Role will fail / Access Denied because user2 does NOT have sts:AssumeRole permission.

